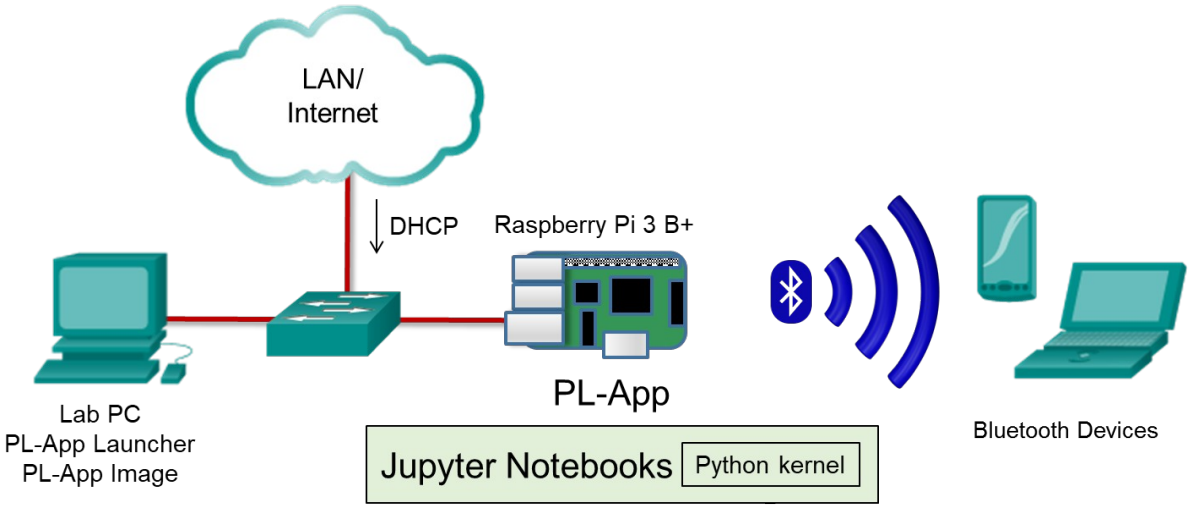
Lab - Sniffing Bluetooth with the Raspberry Pi

Keep Standard Topology



Objectives

In this lab, students will become familiar with varying levels of security on different common devices that use Bluetooth/BLE (Bluetooth Low Energy). Students will configure a Raspberry Pi to detect and display information about the Bluetooth devices that are in its range. This is an awareness building lab that will help students understand the nature of wireless hacking processes and requirements and take on a "hacker mindset" to think like a "white hat" to help secure IoT Security devices.

Part 1: Configuring the Raspberry Pi as a Bluetooth Sniffer

Part 2: Using the CLI to Detect and Display Information about Bluetooth Device

Part 3: Using the CLI to Pair with a Sniffed Device

Background/Scenario

Bluetooth offers several benefits and advantages. Bluetooth also has risks.

Bluetooth technology and associated devices are susceptible to general wireless networking threats, such as denial of service attacks and man in the middle (MITM) attacks. MITM attacks occur when the threat actors position themselves between the source and destination and intercept the communications. Some Bluetooth devices are designed to broadcast MAC, universal unique identifier (UUID) and service information at a predefined interval. Due to continuous advertisement, hackers can easily track the device and decode the broadcasting information using a sniffer. This lab requires that the target Bluetooth devices are locatable by the Raspberry Pi. These could be phones, laptops, personal fitness devices, and others.

Note: The labs in this course assume that the student has all of the necessary hardware to perform them. If you do not have the necessary hardware and wish to complete these labs, you may wish to purchase kits which contain all of the hardware for the challenge labs and additional hardware, which can be used to

complete additional experiments beyond this course. Make sure to read the required resources for each challenge lab to understand what hardware is required.

Required Resources

- Raspberry Pi 3 Model B or later (with PL-App)
- 8GB Micro SD card (minimum required)
- PC with IoTSec Kali VM
- Network connectivity between PC and Raspberry Pi
- Bluetooth devices (PC, Smartphone, Smartband, Bluetooth LED Control…)

Part 1: Configuring the Raspberry Pi as a Bluetooth Sniffer

Step 1: Keep the standard topology.

a. Keep the standard topology with Kali VM and Raspberry Pi connected to your home network.

b. Determine the IP address of your Raspberry Pi as was done in previous labs, if necessary, e.g. using the PL-App-Launcher.

c. You can either use the PL-App-Launcher or directly the web browser of your host computer or the web browser in the Kali VM to access the PL-App on the Raspberry Pi. If you use the Kali VM, log into Kali VM with the username **root** and password **toor**.

Step 2: Verify and activate the Bluetooth interface/services on the Raspberry Pi.

a. Within the PL-App, open a terminal window and enter the **rfkill list** command to check if the Bluetooth interface is blocked by software:

```
(pl-app) root@pi:/home/pi/notebooks# rfkill list
0: phy0: Wireless LAN
        Soft blocked: no
        Hard blocked: no
1: hci0: Bluetooth
        Soft blocked: no
        Hard blocked: no
```

b. If the Bluetooth Interface hci0 is blocked, use the command **rfkill unblock bluetooth** to unlock it.

c. Use the **systemctl status bluetooth.service** command to show the status of the Bluetooth service.

```
(pl-app) root@pi:/home/pi/notebooks# systemctl status bluetooth.service
● bluetooth.service - Bluetooth service
   Loaded: loaded (/lib/systemd/system/bluetooth.service; enabled; vendor
preset: enabled)
   Active: active (running) since Tue 2018-05-22 00:00:02 UTC; 10h ago
     Docs: man:bluetoothd(8)
 Main PID: 522 (bluetoothd)
   Status: "Running"
   CGroup: /system.slice/bluetooth.service
           └─522 /usr/lib/bluetooth/bluetoothd
<output omitted>
```

d. Ignore any errors on the Sap driver and/or server like: **Sap driver initialization failed**.

e. If the Bluetooth service status is not “active (running)”, use the command **systemctl start bluetooth.service** to start it.

```
(pl-app) root@pi:/home/pi/notebooks# systemctl start bluetooth.service
```

f. Use the **hciconfig hci0** command to display the status of the Bluetooth interface.

```
(pl-app) root@pi:/home/pi/notebooks# hciconfig hci0
hci0:   Type: Primary  Bus: UART
        BD Address: B8:27:EB:72:CC:11  ACL MTU: 1021:8  SCO MTU: 64:1
        UP RUNNING
        RX bytes:6770 acl:46 sco:0 events:230 errors:0
        TX bytes:3819 acl:46 sco:0 commands:90 errors:0
```

g. If the hci0 interface status is down, use the command **hciconfig hci0 up** to activate it.

```
(pl-app) root@pi:/home/pi/notebooks# hciconfig hci0 up
```

Part 2: Using the CLI to Detect and Display Information about Bluetooth Devices

a. From the command line, use the command **bluetoothctl** to launch the Bluetooth tool. The prompt will change to **[bluetooth]** and will be displayed. It will display some information about the Bluetooth controller of your Raspberry Pi and probably some information about Bluetooth devices connected or available for pairing:

```
(pl-app) root@pi:/home/pi/notebooks# info
[NEW] Controller B8:27:EB:72:CC:11 raspberrypi [default]
[bluetooth]#
```

b. Enter the **scan on** command to start searching for nearby Bluetooth devices and **scan off** to stop it. You should see the MAC addresses and sometimes the device name of all Bluetooth devices discovered by the Raspberry Pi:

```
[bluetooth]# scan on
Discovery started
[CHG] Controller B8:27:EB:72:CC:11 Discovering: yes
[NEW] Device A4:C1:38:CC:F5:42 Triones-A4C138CCF542
[NEW] Device DA:B6:4A:F2:A8:BA Band
[NEW] Device 28:98:7B:E1:CE:D8 GT-S700
[bluetooth]# scan off
```

Take note of the MAC address (and if available the name) of some discovered Bluetooth devices:

51:F3:FC:10:E6:A9

c. Use the **info** command followed by the MAC address of a discovered device to obtain detailed information about the device:

```
[bluetooth]# info DA:B6:4A:F2:A8:BA
Device DA:B6:4A:F2:A8:BA
        Name: Band
        Alias: Band
        Paired: no
        Trusted: no
```

```
        Blocked: no
        Connected: no
        LegacyPairing: no
        UUID: Anhui Huami Information.. (0000fee0-0000-1000-8000-00805f9b34fb)
        ManufacturerData Key: 0x0157
```

Take note of the information displayed and perform a web search for the meaning of some fields displayed and report below a summary of the main information displayed.

[bluetooth]# info 51:F3:FC:10:E6:A9
Device 51:F3:FC:10:E6:A9 (random)
Alias: 51-F3-FC-10-E6-A9
Paired: no
Trusted: no
Blocked: no
Connected: no
LegacyPairing: no
ManufacturerData Key: 0x004c
ManufacturerData Value:
10 06 0e 1a 40 df a8 f6

Part 3: Using the CLI to Pair with a Sniffed Device

a. Use the command **paired-devices** to view the devices actually paired with the Raspberry Pi.

```
[bluetooth]# paired-devices
```

Report below the MAC address and name of the Bluetooth devices actually paired with your Raspberry Pi.

paired-devices
[DEL] Device 51:F3:FC:10:E6:A9 51-F3-FC-10-E6-A9
[DEL] Device 43:0D:9E:80:41:4F 43-0D-9E-80-41-4F
[DEL] Device 7D:D4:6F:C1:80:B2 7D-D4-6F-C1-80-B2
[DEL] Device 8C:EA:48:64:0A:33 [TV] Samsung AU7179 75 TV

b. Using the command **pair** followed by the MAC address, for example, **pair CA:4C:FB:5E:00:BA**, try to pair via Bluetooth the Raspberry Pi to some previously discovered devices.

```
[bluetooth]# pair CA:4C:FB:5E:00:BA
Attempting to pair with CA:4C:FB:5E:00:BA
<Output omitted>
[CHG] Device CA:4C:FB:5E:00:BA Paired: yes
Pairing successful
```

Are you able to pair without permission from the owner to some Bluetooth devices? Please, report below your findings:

I am note able to connect, without the authorization of the owner
this Bluetooth Device implement a high level of security.

c. Use the command **paired-devices** to view the devices now paired with the Raspberry Pi.

```
[bluetooth]# paired-devices
Device CA:4C:FB:5E:00:BA K800BA
```

Record the MAC address and name of the Bluetooth devices actually paired with your Raspberry Pi.

the same as before

d. Using the command **connect** followed by the MAC address, for example, **connect CA:4C:FB:5E:00:BA**, try to connect via Bluetooth the Raspberry Pi to some previously paired devices.

```
[bluetooth]# connect CA:4C:FB:5E:00:BA
Attempting to connect to CA:4C:FB:5E:00:BA
[CHG] Device CA:4C:FB:5E:00:BA Connected: yes
Connection successful
[CHG] Device CA:4C:FB:5E:00:BA ServicesResolved: yes
[K800BA]#
```

Are you able to connect (without permission from the owner) to some Bluetooth devices? Write your findings below:

I am note able to connect, without the authorization of the owner this Bluetooth Device implement a high level of security.

e. Enter **quit** to exit the Bluetooth tool.

```
[bluetooth]# quit
```

Part 4: Clean Up

a. Enter **systemctl stop bluetooth.service** to disable the Bluetooth service.

```
(pl-app) root@Pi-3:/home/pi/notebooks# systemctl stop bluetooth.service
```

b. Verify that the Bluetooth service has been disabled with the **systemctl status bluetooth.service** command.

```
(pl-app) root@Pi-3:/home/pi/notebooks# systemctl status bluetooth.service
● bluetooth.service - Bluetooth service
   Loaded: loaded (/lib/systemd/system/bluetooth.service; enabled; vendor
preset: enabled)
   Active: inactive (dead) since Tue 2018-11-27 21:47:30 UTC; 1min 10s ago
     Docs: man:bluetoothd(8)
  Process: 843 ExecStart=/usr/lib/bluetooth/bluetoothd (code=exited,
status=0/SUCCESS)
 Main PID: 843 (code=exited, status=0/SUCCESS)
   Status: "Quitting"
<output omitted>
```